Deriving the suvat Formulas

Exam code: 9MA0

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Deriving the suvat equations

What does suvat stand for?

- **suvat** is an acronym for the five quantities used when **modelling motion** in a **straight-line** with **constant acceleration**
  - **s** – displacement (from the starting point)
  - **u** – initial velocity
  - **v** – final velocity
  - **a** – acceleration
  - **t** – time
- All except time are **vector** quantities and can be **negative**
  - time is a **scalar** quantity

What are the suvat equations for constant acceleration?

- The **five** suvat equations for motion in a **straight line** are:

Stuck on something?
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Got it

$$s = \frac{1}{2}(u + v)t$$

$$s = ut + \frac{1}{2}at^2$$

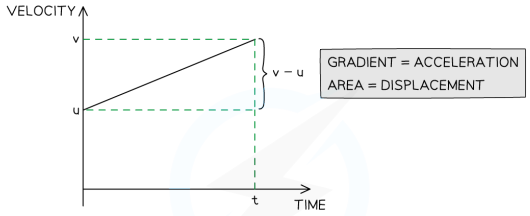
$$s = vt - \frac{1}{2}at^2$$

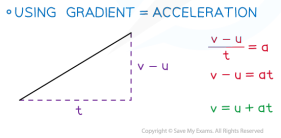
- The equations can **only be used** when the motion has **constant acceleration**
- All equations connect four of the five quantities
  - Knowing any **three** allows a **fourth** to be found

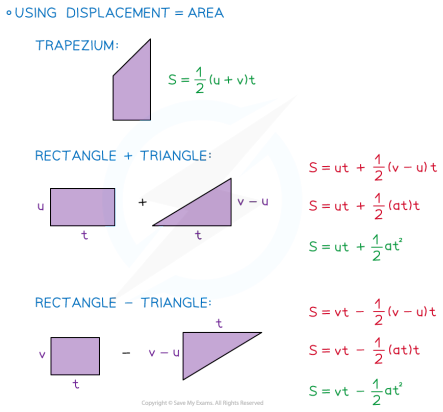
- The equations are **provided** in the exam

How do I derive the suvat equations?

- The **four** equations that involve time can be derived from a **velocity-time** graph
  - The velocity-time graph will be a **straight line** as the **acceleration** is **constant**
  - The fifth equation can be found by choosing any two of the equations and eliminating the t variable (see the worked example)







- **Two** of the equations can also be derived using **calculus**
  - **Velocity** is found by **integrating acceleration**
  - **Displacement** is found by **integrating velocity**

When $t = 0$, velocity $= u$ $\quad u = a(0) + c$

$c = u$

At time t, velocity $= v$ $\quad v = at + u$

$v = u + at$

Displacement $= \int$ (Velocity) dt

Displacement $= \int v\,dt = \int (u + at)\,dt = ut + \frac{1}{2}at^2 + c$

At time $t = 0$, displacement $= 0$ $\quad 0 = u(0) + \frac{1}{2}a(0)^2 + c$

$c = 0$

At time t, dispacement $= s$ $\quad S = ut + \frac{1}{2}at^2$

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Worked Example

Use the constant acceleration equations

$$s = \frac{1}{2}(u + v)t \text{ and } v = u + at$$

to show that

$$v^2 = u^2 + 2as.$$

Answer:

We need to eliminate the 't'

Step 1: Rearrange $v = u + at$

$t = \frac{v - u}{a}$

Step 2: Substitute into $s = \frac{1}{2}(u + v)t$

$S = \frac{1}{2}(u + v)\left(\frac{v - u}{a}\right)$

Step 3: Rearrange into the required form

$2as = (u + v)(v - u)$ by multiplying by $2a$

$2as = v^2 - u^2$ by expanding

$v^2 = u^2 + 2as$

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Examiner Tips and Tricks

- If you are asked to derive one of the formulae then the question will likely give you a hint as to which method to use. They may provide a velocity-time graph. Make sure you show each step and state any reasons such as the gradient of the graph being the acceleration.
- If the question does not ask you to derive the formulae, then you can use them freely without proof.

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